Java
Home Assignment-2

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Aim: To develop a Banking system in java to manage customer accounts using object-oriented programming concepts such as classes, objects, constructors, encapsulation, methods, access specifiers and packages.

Theory:

class: A class is a blueprint that defines the data and methods of an object.

object: An object is an instance of a class used to access its data and methods.

Encapsulation: Encapsulation is the process of binding data and methods together in a class and restricting direct access to data using private members.

constructor: A constructor initialize as object. Java supports default, parameterized and copy constructors.

constructor chaining: calling one constructor from another using this() is called constructor chaining.

Package: A package is used to group related Java classes and organize the program.

classes used:

- Bank - store bank information.
- customer - store customer details.
- Account - stores account details and perform deposit & withdrawl operations.

source code:

Account.java

```
Package banking;
public class Account{
    private int accNo.;
    private string name;
    private double balance;
   //Default constructor.
public Account() {
      this(0, "Unknown", 0);
}
//Parametrical constructor
```

```
Public Account (int accNo, string name, double balan
    this.accNo = accNo;
    this.name = name;
    this.balance = balance;
}
//copy constructor
Public Account (Account a) {
    this(a.accNo, a.name, a.balance);
}
Public void deposit(double amount) {
    balance += amount;
}
public void withdraw(double amount) {
    if (amount <= balance)
        balance -= amount;
}
public void displayAccount() {
    int x=1;
    System.out.println("Account No.:" +accNo);
    System.out.println("Name:" + name);
    System.out.println("Balance: Rs." + balance);
}
public string getName() {
    return name;
}
public void setName(string Name) {
    this.Name = name;
}
}
```

Bank.java

```
Package banking;
public class Bank {
    string bankName;
    static int totalAccounts = 0;
    public Bank(string bankName) {
        this.bankName = bankName;
    }
    public static void displayBankInfo() {
        System.out.println("Bank: ABC Bank");
        System.out.println("Total Accounts:" + total
                                    Accounts);
    }
}
```

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stomer.java

```
Package banking;
Public class customer {
  private int id;
  private string name;
  public customer(int id, string name) {
      this.id = id;
      this.name = name;
  } public void display(customer() {
      system.out.println("customer ID:," +id);
      system.out.println" customer Name:"+ name);
    }
  }
```

AccessDemo.java

```
Package banking;
public class Access Demo {
    public int a=10;
    private int b=20;
    protected int c=30;
    int d=40; // default.
  public void display() {
    system.out.println("Public:"+a);
    system.out.println("Private:"+b);
    system.out.println("Protected:"+c);
    system.out.println("Default:"+d);
  }
 }
```

Main.java

```
important banking.*;
public class main {
    public static void main(string[] args) {
     Bank b = new Bank("ABC Bank");
     customer c= new customer(101, "vennela");
    Account a1 = new Acount(1001, "vennela", 10000);
    Account a2 = new Account();  // Default construct
     Account a3 = new Account(a1); //copy constructo
   Bank.totalAccount = 3;
   Bank.displayBankInfo();
```

```
        c.displayCustomer();
        a1.deposit(5000);
        a1.withdraw(2000);
        a1.displayAccount();
        System.out.println("Name:" +a1.getName());
        a1.setName("Sri Vennela");
        AccessDemo abj = new AccessDemo();
        obj.display();
    }
}
```

Package structure:

```
Banking project
    banking
        Account.java
        Bank.java
        customer.java
        AccessDemo.java
    main.java
```

- sample output:

```
Bank: ABC Bank
Total Accounts: 3
customer ID: 101
customer Name: Vennela
Account No.: 1001
Name: Vennela
Balance: Rs-13000.0
Name: Vennela
public: 10
private: 20
protected: 30
Default: 40
```

conclusion:

The Banking system was successfully implemented using Java oop concepts. It demonstrate classes, objects, constructors, constructor chaining, encapsulation, variables, methods, access specifiers and packages.